

From Mesh to Ornament

Subdivision as a generative system

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Abstract. *This paper explores the use of subdivision algorithms for the production of three-dimensional ornament. In a first step, this paper presents modifications to the Catmull Clark and Doo Sabin processes' weighting schemes. In a second step, it proposes how these modified processes can be applied specifically to the generation of ornament. It presents methods for specifying weights using parameters both intrinsic and extrinsic to the mesh. Strategies for working with both very uniform and more differentiated input meshes are considered.*

Keywords. *Subdivision; ornament; generative system*

Introduction

Subdivision processes have traditionally been used in computer graphics to generate smooth surfaces from a coarser polygonal mesh. This paper explores how these subdivision processes can be reconfigured and applied to produce ornament. As a generative system, subdivision processes have the advantage that they are not additive, and do not require any a priori idealized components to combine and arrange. Rather, they present a purely operations-based geometric approach to generating form. Given the appropriate process parameters, they can produce forms of astounding complexity from even the most undifferentiated input.

The first section of the paper introduces modifications to the subdivision processes in the form of weighting parameters to control vertex interpolation and extrusion. The second section explores various approaches to generating ornament using these

modified schemes. In a first example, a uniform input mesh is subdivided using an extrinsic specification of parameters. In a second example, a more differentiated input mesh is subdivided by specifying parameters according to the mesh's intrinsic attributes. The appendix presents ornamental figures that have been generated using these approaches.

Extension of the Subdivision Schemes

Two of the most established subdivision schemes are the Catmull-Clark process (1974, 1978) and the Doo-Sabin process (1978). Both of these subdivision schemes can be understood by considering their two constituent parts: the topological rules and the weighting rules. The topological rules specify how to obtain the combinatorics of the refined mesh from the combinatorics of the input mesh by generating new vertices, edges and faces. The weighting rules specify how to calculate the positions of these new

vertices based on interpolation between vertices of the input mesh.

Both the Catmull-Clark and Doo-Sabin processes generates smooth and rounded forms when one uses their prescribed weighting rules. By introducing parameters to allow for variations in these weighting rules, non-rounded forms with highly diverse attributes can be produced. These attributes can include concavity and convexity, the appearance of branching, porosity, and fractalization – just to name a few. These attributes can be produced both globally and locally, and they can be combined and arranged to compose ornament.

Described below are extensions to these two schemes' weighting rules. Rather than provide a step-by-step description of the two processes, this paper focuses on the changes to the published processes, assuming a quad mesh as an input. Parameters are introduced that control the position of newly generated vertices as interpolations of the positions of previous-generation vertices. Further parameters are added to allow an extrusion of these newly generated vertices. These parameters are henceforth referred to as weights.

Extended Catmull-Clark Scheme

Initial face midpoints can be extruded along the face's normal vector $\vec{n}_f$ using weight w_f .

$$\vec{F}'_1 = (\vec{P}_1 + \vec{P}_2 + \vec{P}_3 + \vec{P}_4)/4 + \vec{n}_f \cdot w_f \quad (1)$$

Edge midpoints are specified as a weighted interpolation (using weight w_1) of the new face points adjacent to the edge with the edge's endpoints.

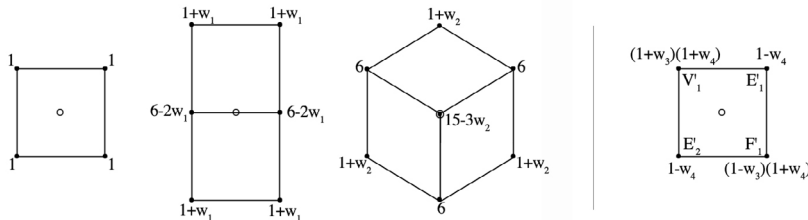


Figure 1
Catmull-Clark stencils for face points, edge points, and vertex points using the introduced weights. The rightmost figure is the face point stencil for use after the first iteration.

These midpoints can be extruded along the edge's normal vector $\vec{n}_e$ (defined as an average of the normal vectors of the adjacent faces) using weight w_e .

$$\vec{E}'_1 = ((\vec{F}_1 + \vec{F}_2)(1 + w_1) + (\vec{P}_1 + \vec{P}_4)(1 - w_1))/4 + \vec{n}_e \cdot w_e \quad (2)$$

Corner points are specified as a weighted interpolation (using weight w_2) of the average $\vec{F}$ of all i face points touching the vertex with the average $\vec{E}$ of all edge midpoints for edges touching the vertex. These new corner points can be extruded along the previous corner point's normal vector $\vec{n}_p$ using weight w_p .

$$\vec{C}'_1 = (\vec{F}(1 + w_2) + 2\vec{E}(1 - w_2)/2) + (\vec{i} - 3)\vec{C}_1/i + \vec{n}_c \cdot w_c \quad (3)$$

After one iteration of this algorithm, vertices produced can be distinguished as having been generated as face midpoints, edge midpoints, or corner points. Thus two additional weights, w_3 and w_4 , can be added to equation (1) to control the placement of subsequent midpoints.

$$\vec{F}'_1 = ((\vec{V}'_1(1 + w_3) + \vec{F}'_1(1 - w_3))(1 + w_4) + (\vec{E}'_1 + \vec{E}'_2)(1 - w_4))/4 + \vec{n}_f \cdot w_f \quad (4)$$

Weights introduced in equations (1) to (4) can be integrated into weighting stencils that depict the interpolation between previous-generation vertices:

Extended Doo-Sabin Scheme

Weight w_1 is introduced to control the interpolation between the vertices of the face. New points can be extruded along the face's normal vector using weight w_f . Equations (5) and (6) refer to faces with 4

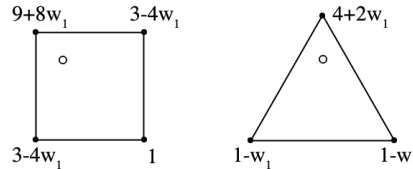
and 3 vertices respectively.

$$P'_1 = (P_1 \cdot (2.25 + 2w_1) + (P_2 + P_3) \cdot (.75 - w_1) + .25P_3)/4 + \vec{n}_f \cdot \vec{w}_f \quad (5)$$

$$P'_1 = \frac{2}{3}P_1(1 + w_1/2) + \frac{1}{6}(P_2 + P_3)(1 - w_1) + \vec{n}_f \cdot \vec{w}_f \quad (6)$$

After one iteration of the algorithm, each new face can be distinguished as deriving from an original face, from connected edges, or from connected vertices. Distinct weights analogous to both w_1 and w_f can thus be introduced for these three groups of faces in subsequent generations.

Figure 2
Modified Doo-Sabin stencils with variable weights, for quads and triangles respectively.



Ornamentation Strategies

On a very general level, the ornamental nature of the forms produced depends on two factors: the topology and topography of the input mesh, and the manner in which specific weighting values are applied to the mesh throughout the subdivision process. It is beyond the scope of this paper to outline which input mesh, combined with which exact weighting values, will produce which distinct ornamental attribute. Rather, this paper will consider possible degrees of differentiation of an input mesh, and outline methods of applying weights in very specific contexts.

Uniform Versus Differentiated Input Meshes

In selecting an input mesh, the question arises whether to employ a simple uniform mesh and uniquely use subdivision processes to manipulate this mesh, or alternatively to employ a differentiated

mesh that already contains information about the form to be generated. As the latter can contain significant topographical information, this information does not need to be conveyed through parameters and generated through the process itself. Fewer non-uniform and non-stationary weights are required to generate appealing results. Therefore a more differentiated mesh can lead to more foreseeable results, and by extension the division process can be easier to control. In addition their geometry is potentially more discernible in the output, making them particularly suitable for generating entire families of forms.

One disadvantage of a differentiated mesh is that its geometry can be confining. In many cases, ornamental attributes appears to be created within the faces of the input mesh, possibly leading to a cellular appearance with frequent repetition of motifs. Forms may appear as if they are made of several pieces or components, rather than being perceived as holistic.

Beyond the advantages and disadvantages of each method, a key question in selecting the input mesh is whether one understands ornament to be an object in itself, or as something adorning or embellishing another object. In the former case, a uniform input mesh offers the highest degree of freedom in generating specific formal characteristics. In the latter case, a mesh as a geometric abstraction of the object to be adorned may be preferable as topographical information is conveyed and the process will be more transparent, traceable and easier to control.

Non-stationary, Non-uniform Weights

Regardless of the degree of differentiation of the input mesh, key to increasing the scope of conceivable forms is the concept of non-stationary weights. This implies that weights can assume distinct values at each iteration of the subdivision process. Certain characteristics of the forms can be generated by single iteration of the process, while others emerge through the interplay of several iterations with distinct parameters. The scope of realizable forms can be further expanded by combining the

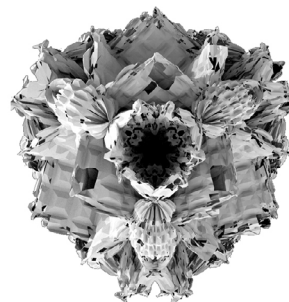
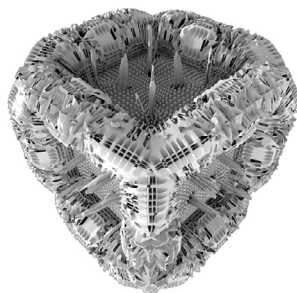
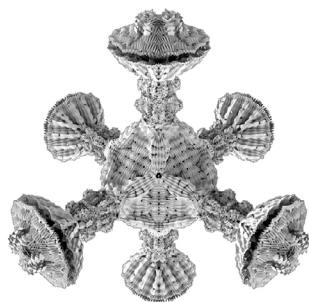


Figure 3
Three cubes after eight subdivision iterations using non-stationary weights. The left and center figures are formed using the expanded Catmull-Clark process. The figure at right is formed using a combination of Catmull-Clark and Doo-Sabin processes.

Catmull-Clark and Doo-Sabin schemes, as the output mesh from any of the two processes can serve as the input mesh for the other. Figure 3 illustrates a very simple uniform input mesh – a cube – that has been subdivided using the expanded schemes with non-stationary weights.

The forms in figure 3 retain the three-fold symmetry of their input mesh because uniform weighting values are applied to each point. By applying distinct weighting values to different parts of the mesh – what is referred to as non-uniform weights – specific local conditions can be generated, and the breadth of characteristics that can be produced increases considerably.

The question arises by what means these non-uniform weights can be specified. Two broad

strategies are considered. The first is to specify parameters in the mesh's environment, i.e. parameters that are extrinsic to the mesh. The second approach is to use the mesh's intrinsic attributes – such as its topology and topography– to specify the weights.

Closely related to the chosen approach is the question of the degree of differentiation of the input mesh that was addressed in the previous paragraph. A uniform input mesh favors an extrinsic specification of weights – particularly in the first iterations of the process – as there are by definition insufficient intrinsic attributes. A more differentiated input mesh allows for both extrinsic and intrinsic specification. Below are two approaches to generating an ornamented column. The first uses a uniform mesh with

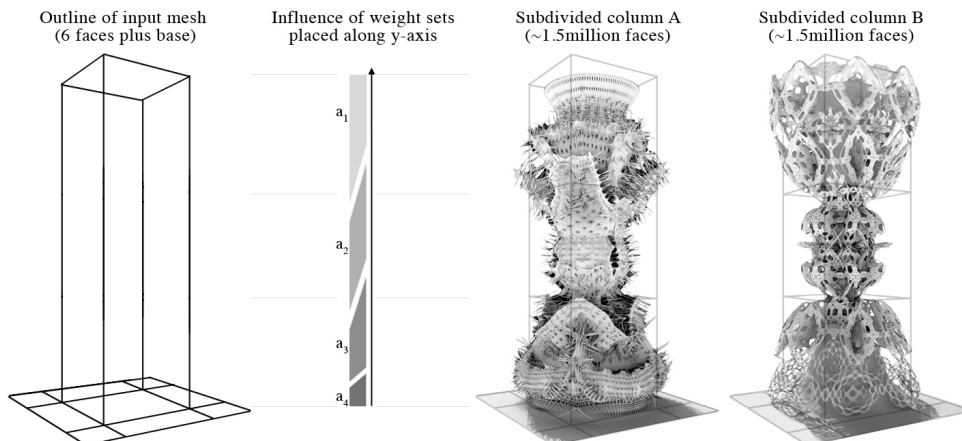


Figure 4
Uniform input mesh, consisting of an elongated cube and a base, with four distinct sets of non-stationary weights applied. Weights are positioned along the y-axis, leading to forms that are differentiated along this axis.

specification of extrinsic parameters, while the second uses a more differentiated mesh with specification according to intrinsic attributes.

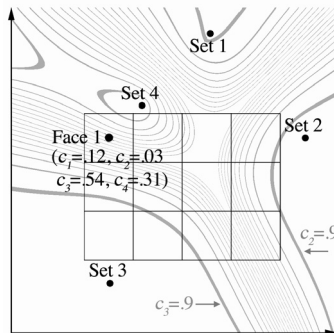
Uniform Input Mesh and Specification of Extrinsic Parameters.

If one views the mesh as being located in an environment, then one can place points that represent sets of subdivision weights in this environment, either at locations directly on a mesh or in its proximity. For each face of the mesh, the distances to these sets are calculated to determine the sets' level of influence and calculate the weighting values used to subdivide the face. Thus scaling, translation and rotation of the input mesh or the environment modifies the position of vertices relative to the weight sets and leads to changes in output.

In a basic version of this scheme, sets of weights are simply placed along an axis or a vector. This concept, using four sets of non-stationary weights positioned along the y-axis, is shown in figure 4.

In both columns depicted in figure 4, the position of the weight sets along the y-axis remains visible: the columns appear to be divided into three broad sections and a base. A gradient between the weights' domains ensures smooth transitions in the topographies and the attributes that these weights produce in the columns. As the input mesh was limited to a uniform elongated cube, eight subdivision iterations were required to generate these columns in their level of detail.

Figure 5
2D example showing the effect of four weights on one face of a 4x3 mesh. The curved lines represent the weight sets' level of influence c , with the bold lines indicating a weighted value of 0.9.



The positions of the weight sets do not need to be confined to a vector. Rather, they can be freely distributed in a plane, as shown in figure 5, or in three-dimensional space.

When positioning weights sets in space, the level of influence of weight c_a of a set a can be calculated as shown in equation (8), where the dist function represents the distance between a set and the face, h is the strength of the set, n is the total number of sets, t is a 'tightness' control exponent greater than 1, and the maximum possible distance of two points is assumed to be 1. The subsequent calculation of a weighting value w_x based on values contained in the sets $w_{x,a}$ and the sets' level of influence weights c is shown in equation (9).

$$c_a = \frac{(1 - \text{dist}(\text{pos}_{\text{face}}, \text{pos}_{\text{set},a}))^t \cdot h_a}{\sum_{i=1}^n (1 - \text{dist}(\text{pos}_{\text{face}}, \text{pos}_{\text{set},i}))^t \cdot h_i} \quad (8)$$

$$w_x = \sum_{i=1}^{n_a} (w_{x,i} \cdot c_i) \quad (9)$$

Differentiated Input Mesh Using Intrinsic Parameters

The more differentiated the input mesh, the more the mesh itself contains a variety of attributes that allow for a discriminatory specification of weights. These attributes can be both on a topographical and topological level. Three methods of specifying weights according to intrinsic parameters are proposed:

1. Specification according to topological motifs
2. Specification according to topological distances
3. Specification according to topographical curvature

Specification according to topological motifs implies classifying the mesh's vertices according to their valence and number of incident faces while taking into account possible face-edge combinations. A series of such common motifs is shown in figure 6.

One can allot specific sets of weights to vertices



based on these motifs. Alternatively one can assign an attractor/deflector value that can be incorporated into the Catmull-Clark subdivision equations. The attractor/deflector value can be combined with two new weights to influence the placement of subsequent face points and edge points. The two equations below are applied in the subdivision process after equations (1) and (2). U is the attractor/deflector value of a point based on its classification, and n is the number of points of a face.

$$F'_{Attracted} = F'_1 + w_6 \cdot \sum_{i=1}^n (P_i - F'_1)u_{P_i} \quad (10)$$

$$E'_{Attracted} = E'_1 + w_7 \cdot ((P_1 - E'_1)u_{P_1} + (P_2 - E'_1)u_{P_2}) \quad (11)$$

An example of this approach is shown in figure 7. A differentiated input mesh contains 3 types of motifs that are used to specify attractor/deflector values which compress or expand certain parts of the generated columns.

By default, the Catmull-Clark process usually

produces only two motifs beyond those contained in the input mesh. These two motifs are the second and third examples from the left in Figure 6, corresponding to the edge points and face points that are generated. However, further motifs can be generated depending on the mesh model used in the subdivision process. Thus, if two vertices produced in the exact same position are treated as a single vertex, then the topology of the mesh changes since this single vertex obtains the edges from both vertices that were in its place. Using this type of mesh model, the subdivision process can produce motifs that were not contained in the input mesh, and for which additional attractor/deflector values can be specified. These constellations can be intentionally invoked by specifying a minimum distance beneath which proximate vertices are joined. Allowing joined vertices introduces non-linearity into the subdivision process: small changes to a weight's value can lead to seemingly abrupt changes in output as the mesh's topology is altered. This is particularly true when limiting the joined vertex's maximum valence: as edges and corresponding faces cannot form, the mesh assumes a distinct porosity.

Specification according to topological distances implies measuring each vertex's distance – in terms

Figure 6
Sample of commonly occurring motifs based on a vertex's valence and incident faces. Note that the fourth and the seventh figures from the left both have 3 faces and 4 edges, yet depict different topologies.

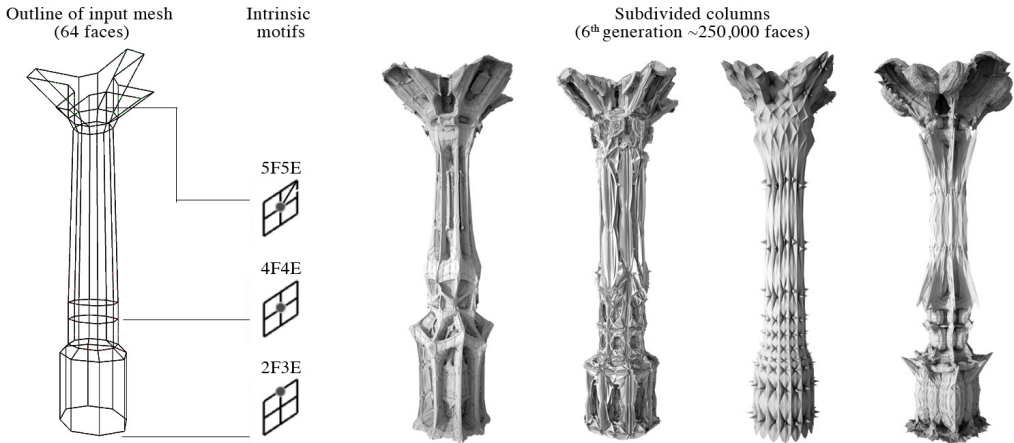
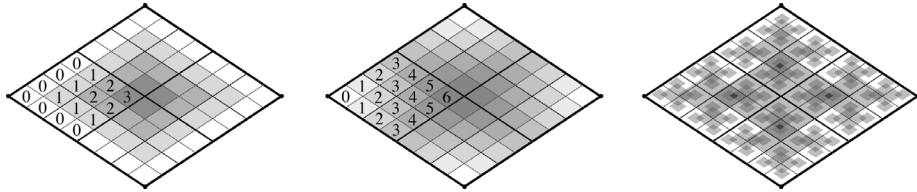


Figure 7
Differentiated input mesh with 64 faces in which the topography and topology approximates a column. Attractor/deflector values are assigned according to three motifs that are intrinsic to the input mesh.

Figure 8

Left and center figures are 3rd generation subdivisions of a quad showing the distance faces to original edges and original vertices respectively. Right figure is a 5th generation subdivision of quad with shading to differentiate whether faces are derived from vertex or corner points.



of number of edges that must be traversed - to an original edge or original vertex. For each weight in equations (1) to (4), one can, for instance, specify two sub-values (corresponding to the minimum and maximum distances) and interpolate between them depending on the distance. This concept is shown in figure 8.

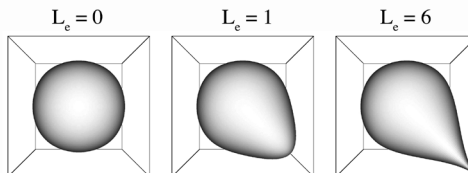
Specification according to topographical curvature implies measuring the local curvature at each face of the mesh, for instance by calculating the face's planarity. The planarity value can be used to amplify or attenuate the vertex weights in equations (1) to (4). Alternatively one can specify sub-values for the weights and interpolate between them based on planarity value.

Further Specification Options

A further method to specify non-uniform weights, which is suitable for both uniform and differentiated meshes, is through tagging of vertices and faces in the input mesh. Specific vertices can be assigned to vertex groups, or faces can be assigned to face groups. Sets of weights can then be specified per group. This approach also enables the locking of certain parts of the input mesh, by fixing the positions of vertices belonging to a certain group for one or multiple iterations of the subdivision process. Locking parts of the mesh, as shown in figure 9, allows for the generation of hard edges, creases, and spikes.

Figure 9

Seventh-generation subdivision of a cube with the bottom right edge of its input mesh locked, where L_e specifies the number of iterations during which the edge remains locked.



It also affords a degree of control over curvature. A similar scheme was used in the animated film *Geri's Game*, as presented by DeRose, Katz and Truong (1998).

Conclusion

By adding degrees of freedom to established subdivision processes, the scope of forms that can be produced increases greatly. In prescribing precisely the manner in which these processes are applied to a mesh, very specific local attributes can be generated. The interplay of these processes over several generations can be manipulated to create ornamental form that exhibits an astounding degree of complexity.

Unlike ornament generated through typical additive processes, these forms are not explicable through reductionism. It is, on the contrary, often difficult to discern the source or input of the forms, much less the exact nature and parameters of the processes applied. Despite this non-reductionism, the processes themselves behave mostly linearly. Small changes in weighting values lead to gradual, traceable changes in output. Though not entirely predictable, the processes are deterministic and they are reproducible.

Further work could begin to classify and delineate combinations of process parameter and the geometric attributes they produce. Beyond this, one could explore the use of geometric constraints on the level of the intermediary meshes to gain an additional degree of control. This could allow subdivision processes to decisively transcend their role as smoothing algorithms and to establish their use as generative systems.



Figure 10
Five columns generated using four sets of vertically aligned non-uniform, non-stationary weights. Top columns are similar to those in figure 4, yet two additional subdivision iterations were performed.

Figure 11
Family of columns based on
a single 64-face input mesh.
Local topographical motifs
are used to specify sets of
non-uniform, non-stationary
weights.



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